

Replication Assignment - Smets and Wouters (2007) Under the No-Growth Assumption

Modification by imposing a unit gross growth factor ($\gamma = 1$) and removing the deterministic trend in technology that drives long-run growth in the original specification transforms the growth stationary Smets and Wouters framework into a fully stationary model. The output, consumption, investment, and wages fluctuate around constant steady-state levels.

For no growth assumption to be internally consistent, the measurement equations were rewritten as pure first differences for real variables,

$$dy = y - y(-1), \quad dc = c - c(-1), \quad dinve = i - i(-1), \quad dw = w - w(-1).$$

Inflation, nominal interest rate, and hours are observed directly,

$$pinfobs = pinf, \quad robs = r, \quad labobs = l.$$

Dynare's `prefilter = 1` option is used for estimation to demean the data, and the structural parameters. Shock standard deviations, and AR coefficients are then estimated following the Bayesian specification in the `.mod` file.

The dynamic impulse responses generated by the estimated model remained qualitatively similar to those in the original study even though there are differences in steady-state scaling. A positive technology shock increases output, consumption, and real wages while lowering inflation. A marginal investment efficiency shock generates a strong short-run increase in investment. A monetary tightening (shock) leads to a contraction in real activity and inflation. However, the magnitudes of these responses are quantitatively smaller and less persistent compared to baseline Smets–Wouters.

Given no growth assumption, technology shocks no longer shift the long-run growth path. This reduces the propagation of shocks through capital accumulation and consumption smoothing. Inflation and wage responses also revert faster to the mean.

We tried to closely follow the original procedure and framework. Our estimation block applies mode finding, posterior sampling (MCMC), and Bayesian impulse responses via `bayesian_irf`. To compute first order stochastic simulations we used `stoch_simul` which generated posterior mean IRFs. The core short-run propagation mechanisms of Smets and Wouters (2007) remain intact in all three shocks despite the absence of deterministic growth reduces the long-run dynamic effects.

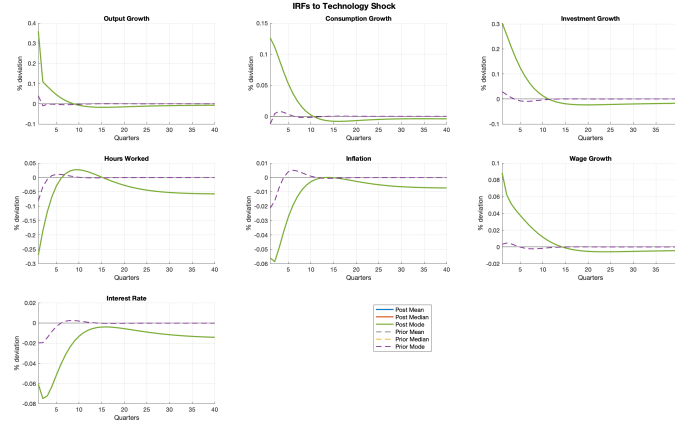
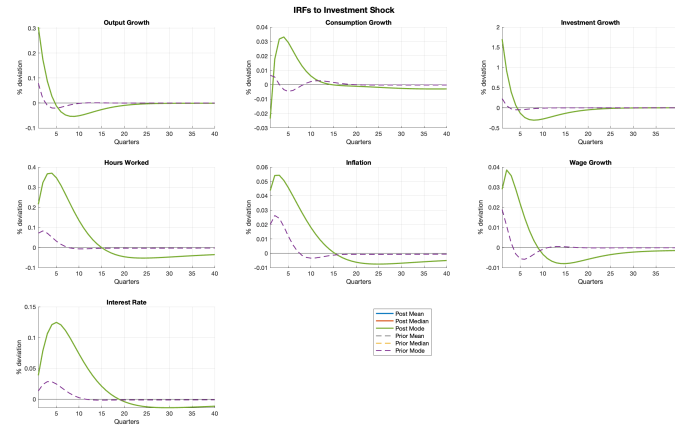
In conclusion, the no growth version of the replication seems to deliver a empirically sound variant of the Smets–Wouters model. Impulse responses differ in magnitude from the originals, however the qualitative business cycle dynamics are visible. This exercise indicates that the key short-run mechanisms in Smets and Wouters (2007) do not critically depend on the presence of deterministic trend growth.

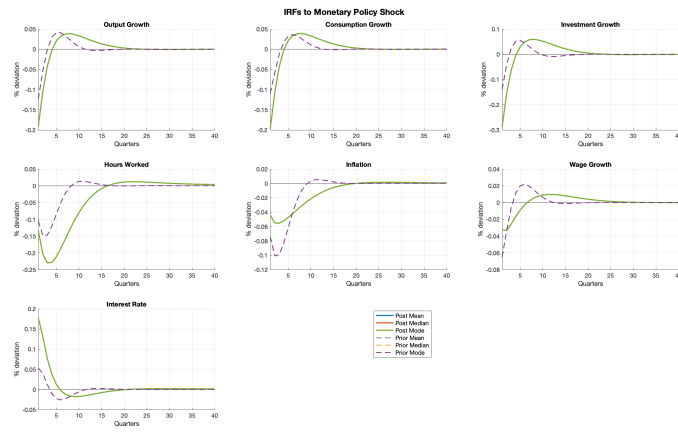
Table 1A: Structural and Shock Process Parameters (No-Growth Version)

Parameter	Mean	Std	HPD 5%	HPD 95%
phi	5.4124	1.0152	3.7452	7.0231
sigma_c	1.6873	0.1982	1.3694	2.0077
lambda	0.6607	0.0467	0.5866	0.7386
xi_w	0.7283	0.0616	0.6309	0.8364
sigma_l	1.7104	0.5789	0.7363	2.5931
xi_p	0.6266	0.0436	0.5526	0.6964
iota_w	0.5604	0.1221	0.3680	0.7632
iota_p	0.2766	0.0969	0.1094	0.4263
psi	0.6121	0.1058	0.4303	0.7754
phi_p	1.5900	0.0772	1.4660	1.7158
r_pi	2.0134	0.1628	1.7454	2.2769
rho	0.8203	0.0223	0.7822	0.8549
r_y	0.1088	0.0254	0.0682	0.1511
r_dy	0.2277	0.0278	0.1828	0.2750
pi_bar	0.6408	0.0992	0.4839	0.7931
beta_const	0.2813	0.1086	0.1124	0.4602
l_bar	0.0946	2.0090	-3.2000	3.4915
alpha	0.1966	0.0172	0.1679	0.2234
rho_a	0.9758	0.0072	0.9645	0.9871
rho_b	0.2235	0.0891	0.0858	0.3647
rho_g	0.9204	0.0238	0.8812	0.9575
rho_i	0.7145	0.0577	0.6207	0.8103
rho_r	0.1432	0.0635	0.0410	0.2434
rho_p	0.8277	0.0548	0.7434	0.9174
rho_w	0.9437	0.0207	0.9086	0.9750
Mu_p	0.6004	0.1312	0.3908	0.7992
Mu_w	0.8402	0.0558	0.7515	0.9289
rho_ga	0.5216	0.0786	0.3957	0.6574

Table 1B: Standard Deviations of Structural Shocks

Shock	Mean	Std	HPD 5%	HPD 95%
eta_a	0.4564	0.0289	0.4084	0.5027
eta_b	0.2403	0.0248	0.1977	0.2796
eta_g	0.5055	0.0265	0.4618	0.5476
eta_i	0.4707	0.0479	0.3903	0.5462
eta_r	0.2407	0.0151	0.2158	0.2628
eta_p	0.1428	0.0168	0.1152	0.1701
eta_w	0.2393	0.0207	0.2054	0.2732

Figure 1: Bayesian IRF: Technology Shock (η_a)Figure 2: Bayesian IRF: MIE shock (η_i)

Figure 3: Bayesian IRF: Monetary Policy Shock (η_r)